



ScleroDx - A Diagnostic Algorithm WebTool

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 Scleroderma Prediction & Test Recommendation

 Patient Symptoms ⓘ

Describe patient symptoms...

 Clinical Features ⓘ

☐ Swollen joints

☐ Myalgia

☐ Dyspnea (NYHA-stage)

☐ DAS 28 (CRP, calculated)

☐ BNP (pg/ml)

☐ FVC (% predicted)

☐ Sclerodactyly

☐ Muscle weakness

☐ DAS 28 (ESR, calculated)

☐ Body weight (kg)

☐ NTproBNP (pg/ml)

☐ DLCO/SB (% predicted)

Prediction Result

Likely Scleroderma

Scleroderma Probability: 53.6%

Top 1 Recommended Test: Skin thickening of the whole finger distal to MCP (Sclerodactyly)

Top 3 Recommended Tests: Skin thickening of the whole finger distal to MCP (Sclerodactyly), Skin thickening of the fingers of both hands extending proximal to the MCP joints, Conduction blocks

Recommended Antibody Tests

- ANA
General screening for autoimmune disease

Most Likely Subtype: 1

Contents

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2. Current diagnostic pathway
3. How ScleroDx can help
4. How it works
5. Demo
6. Questions?

The Challenge:

- **Rare, overlooked disease:** ~20,000 patients in the UK, with early symptoms often misattributed
- **Devastating delays:** Average 2.4 years from first symptom to diagnosis
- **Irreversible consequences:** Late diagnosis leads to organ damage, increased mortality (3.5x higher than general population)
- **Economic burden:** £82M/year to NHS, £270M to UK economy

Why This Happens:

- Non-specific early symptoms (fatigue, reflux, Raynaud's)
- GPs' limited exposure to rare diseases
- No embedded decision support in GP systems

Market Gaps:

Current Healthcare System Failures

Missing Tools:

- No integrated SSc risk detection in GP software
- Generic symptom checkers inadequate for rare diseases
- Static guidelines insufficient for early detection

Workflow Problems:

- EHR systems lack predictive analytics for rare conditions
- Missed opportunities for timely tests/referrals
- Inaccurate referrals strain specialist clinics

Result: High-risk patients missed, unnecessary referrals for others

Health Equity:

- Females make up 84% of those with Scleroderma, but their time from Raynaud's onset to diagnosis was double that of males
(<https://www.clinexprheumatol.org/abstract.asp?a=7030>)
- African Americans and Asians are more likely to have Scleroderma, are more likely to have severe symptoms, and have the worst treatment outcomes
(<https://shmabstracts.org/abstract/racial-disparities-in-systemic-sclerosis-short-term-outcomes-study/>)

Our Solution - ScleroDx

Core Features:

- **Real-time risk scoring:** Calculates SSc probability using symptoms, history, test results
- **Smart test recommendations:** Suggests optimal next diagnostic step based on information gain
- **Seamless integration:** Embedded directly in GP EHR systems

How It Works:

- Alerts GP when patient profile suggests elevated SSc risk
- Provides expert-level diagnostic vigilance at every consultation
- Guides optimal testing to avoid trial-and-error approaches

Value Proposition

MVP vs MVI Strategy

Minimum Viable Product (MVP):

- Focused decision-support plugin for GP software
- SSc probability flagging with actionable test recommendations
- Trained on thousands of real patient cases
- Seamless EMR integration

Maximum Value Impact (MVI):

- Expand to other rare autoimmune diseases
- Incorporate wearable data and advanced imaging
- Become comprehensive "Rare Disease Radar" platform
- Continuous AI model improvements

Target Users & Stakeholders

Multi-Stakeholder Impact

Primary Users:

- **GPs:** Enhanced diagnostic confidence and clear guidance
- **Patients:** Faster diagnosis, better outcomes, reduced anxiety

Key Stakeholders:

- **Specialists:** More accurate referrals, efficient clinic time
- **GP Practices:** Improved care quality metrics
- **Healthcare Systems:** Reduced costs, better value-based care
- **EHR Vendors:** Enhanced product capabilities and differentiation

Market Analysis

Significant Revenue Opportunity

Total Addressable Market (TAM):

- Global: £330-400M today → £530M by 2030
- 3.3M GPs worldwide × £100 license

Serviceable Available Market (SAM):

- Europe + UK: £52M today → £85M in 5 years
- 520k GPs × £100 annual license

Serviceable Obtainable Market (SOM):

- UK focus: £3.9M maximum potential
- Realistic target: £0.5-0.7M ARR by Year 5 (12-15% market share)

Business Model

Scalable SaaS Revenue Strategy

Primary Revenue Stream:

- **B2B Licensing to EHR Vendors** (EMIS, SystmOne, Vision)
- Recurring license fees per user/practice
- Revenue sharing with vendor partners

Value-Based Pricing:

- Health economics justification
- Cost savings from reduced misdiagnoses
- NHS procurement and insurance reimbursement potential

Expansion Strategy:

- Platform play: Add new rare disease modules
- High-margin, scalable growth model
- Global rollout through major primary care platforms

Technical Implementation

Model Development:

- **Logistic Regression:** 78% accuracy, optimized for clinical relevance
- **Random Forest:** Enhanced feature importance analysis
- **Advanced RNN:** 79% test accuracy with time-series analysis

Key Clinical Features:

- Integrated bias mitigation, including female pro-weighting
- Autoantibody test recommendations (ANA, SSc-specific antibodies)
- Further test suggestions to confirm or rule out Scleroderma, significantly reducing diagnosis wait times

Integration Ready:

- Frontend: <https://scleroderma-api.onrender.com/>
- Backend: Deployed and tested
- GitHub: <https://github.com/Amelia3141/Hackathon>

Logistic Regression Insights:

- Suitable to model the binary outcome
(Scleroderma present: Yes, No)
- Model evaluated associations between symptoms and likelihood of Scleroderma.

Logistic Regression (patient_id dataset):

- Most Features (Symptoms) had negative coefficients
- Hence reduced likelihood of Scleroderma predictive value

Top Predictors of Scleroderma

- Skin thickening distal to MCP (Sclerodactyly) → Coefficient: +0.34
- Skin thickening proximal to MCP (both hands) → Coefficient: +0.26

Key takeaway:

- Diagnosis relies on a specific combination of features, not isolated symptoms.

Logistic Regression (patient_id dataset):

Accuracy: 0.79 (Misleading due to Class imbalance)

Model failed to detect any true positives (Scleroderma = 1):

- Precision = 0.00, Recall = 0.00, F1 = 0.00

Excellent performance for the majority class (No Scleroderma):

- Precision = 0.79, Recall = 1.00, F1 = 0.88

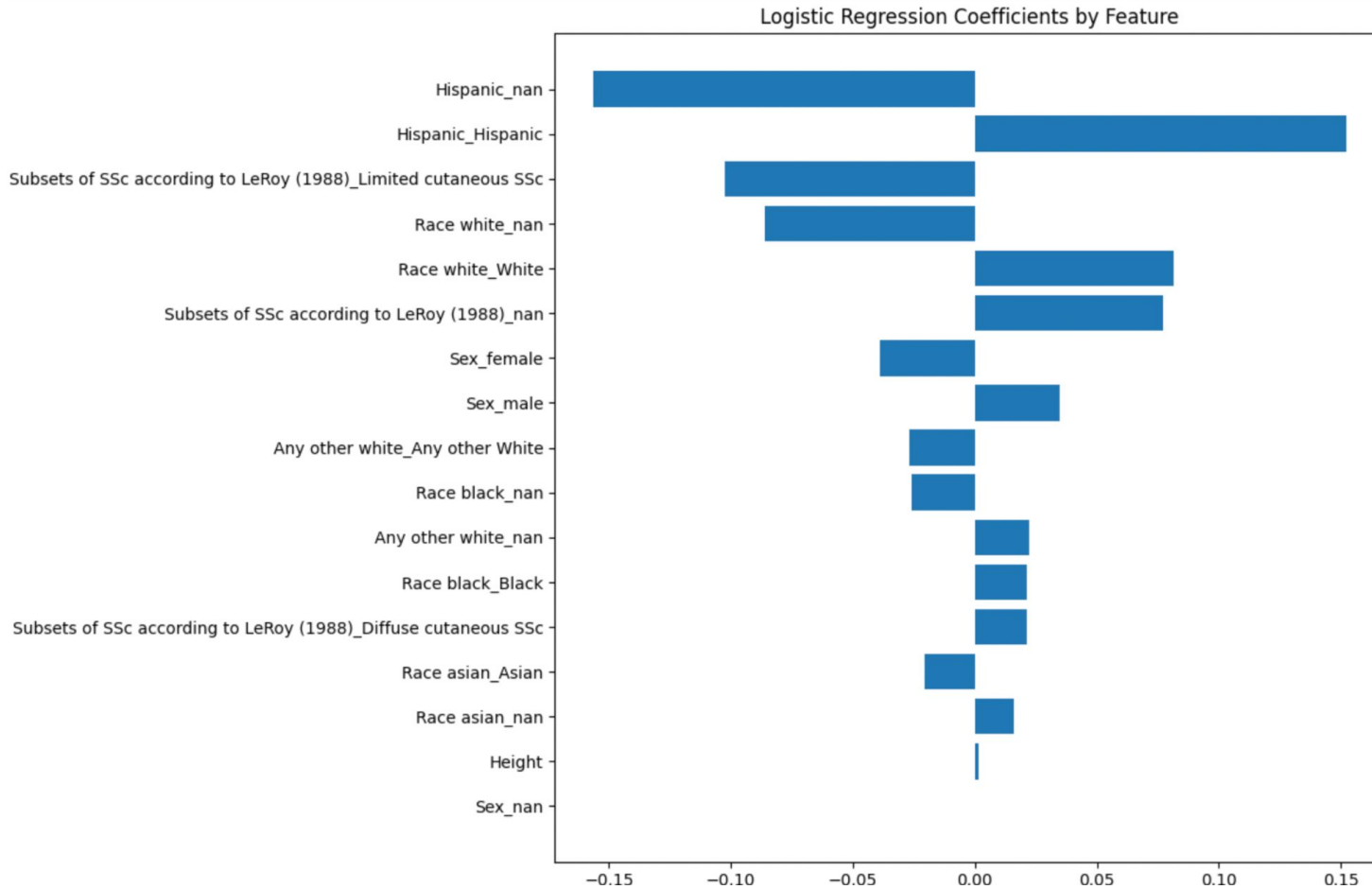
Logistic Regression (patient_id dataset):

- Most Features (Symptoms) had negative coefficients:
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Top Predictors of Scleroderma

- Hispanic_Hispanic → Coefficient: + 0.152217
- Subsets of SSc_Limited cutaneous SSc → Coefficient: + 0.076893
- Race white_White → Coefficient: + 0.081629

Coefficient Visualisation (pats dataset):



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Random Forest

Using 10 key clinical variables

Heavily biased towards most classes, almost impossible to identify positive classes

Using XGBoost

Train XGBoost after using SMOTE.

Added important variables such as Age, SSc subtype, HRCT, etc.
Add PR curve evaluation

Top two important features:
skin thickness (proximal to MCP & sclerodactyly)

Random Forest

XGBoost

Voting Integration

**add more medical
variables + SMOTE
resampling +
threshold adjustment**

RF + XGB

Using VotingClassifier (soft voting)
Combining the first two model predictions
Use all fine screening features (15+)

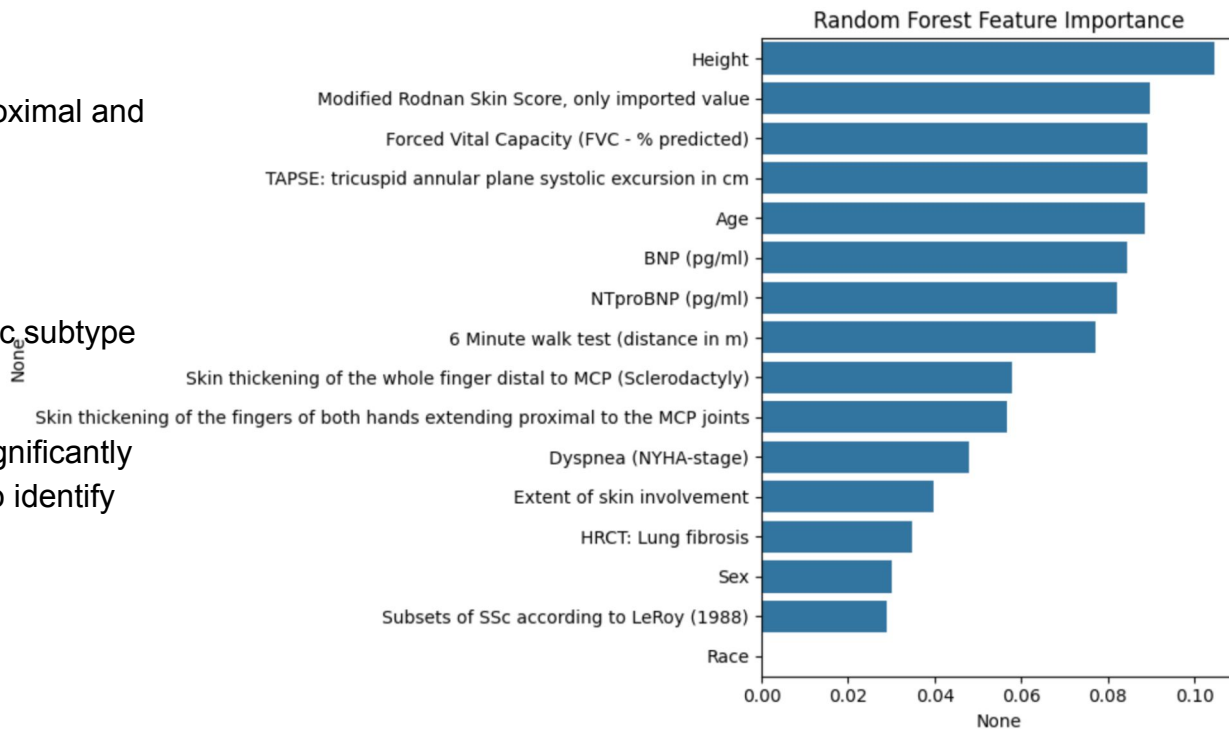
Accuracy: 0.59
Recall (Class 1): 0.57
Precision (Class 1): 0.28
ROC-AUC: 0.613

Feature Importance - RF

Top five important features include:

1. Skin thickening of fingers (proximal and distal to MCP)
2. Sex (higher risk in females)
3. HRCT: Lung fibrosis
4. Dyspnea
5. LeRoy subclass (used for SSc subtype diagnosis)

These clinical features contribute significantly to the classification model's ability to identify systemic sclerosis.

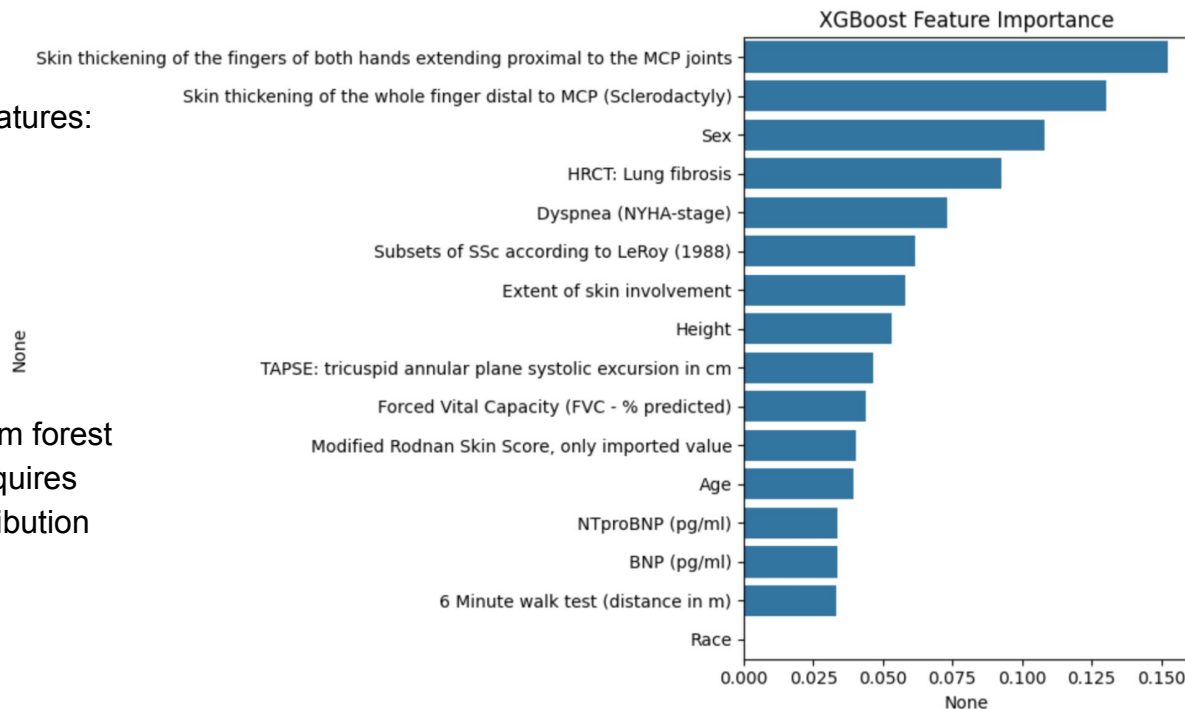


Feature Importance - XGB

Slightly different ranking, top three features:

1. Height
2. Modified Rodnan Skin Score
3. FVC (lung function)

The high ranking of "Height" in random forest may indicate bias or data issues—requires investigation of potential feature distribution imbalance.



GARNN Time-Series Analysis

- **Graph Attention Networks (GAT/GATv2) with Gated Recurrent Units (GRU) to model multivariate time series data for disease prediction.**
- **Addresses the "black box" problem common in traditional deep learning approaches by providing interpretable insights into temporal variable importance**
- **Graph Attention: Graph neural network layers (GATConv) to capture complex relationships between features (e.g., lab values, symptoms) and between patients.**
- **Recurrent Neural Network: GRU (Gated Recurrent Unit) to process sequential or longitudinal data for each patient, allowing the model to learn from time-series and repeated measurements.**
- **Fully Connected Layer: After graph and recurrent processing, a final layer predicts the probability of scleroderma.**

GARNN Time-Series Analysis

Performance Metrics (Best Configuration)

From your latest training run (POS_WEIGHT=10, threshold=0.2):

- **Test Accuracy: 0.793 (79.3%)**
- **Best Threshold: 0.20**
- **Positive Predictive Value (PPV): 0.407**
- **Sensitivity (Recall): 0.349**
- **F1 Score: 0.376**

Other configs:

- **At POS_WEIGHT=2: Sensitivity up to 0.746 but lower PPV and F1.**
- **At POS_WEIGHT=20: Sensitivity up to 0.889 at the cost of much lower PPV.**

Interpretation:

- **The model is tuned for a balance between sensitivity (recall) and precision (PPV), but the threshold and class weights can be adjusted and to favor one over the other depending on clinical needs.**

GARNN future improvements?

1. Data Quality & Quantity:

- Collect more labeled data, especially positive scleroderma cases.
- Improve feature completeness and reduce missing data.

2. Model Enhancements:

- Tune hyperparameters (hidden size, number of layers, learning rate, etc.).
- Try more advanced graph neural network architectures or combine with transformers.
- Incorporate additional patient relationships (e.g., demographic similarity, comorbidities).

3. Clinical Features:

- Add more granular or longitudinal patient data (e.g., time-series lab values).
- Incorporate structured and unstructured data (e.g., clinical notes, imaging).

Model Access:

1. **Input:** Patient data (clinical features, lab values, optionally free-text symptoms).
 2. **Feature Graph:** The model builds a graph where nodes represent features and edges represent relationships between them (e.g., medical relevance).
 3. **Prediction:** Outputs the probability that a patient has scleroderma, along with recommended tests and reasoning (using SHAP with a fallback to Random Forest for explainability)
- Frontend: <https://scleroderma-api.onrender.com/>
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